

Project Executable Files

Date: 15 July 2026

Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / SQL files	Yes
5	Environment configuration file (.env.example)	Yes
6	Deployed application URL (if hosted)	No (or Yes, if deployed)
7	Executable Binary / APK	NA (Web Application)
8	Dockerfile / Containerization Configuration	No (Optional)
9	Test files and test reports	Yes
10	Demo Video / Walkthrough	Yes

Step 2: File / Folder Structure

```
HDI_Prediction_System/  
|  
|— app.py  
|— requirements.txt  
|— README.md
```

```
|— .env.example
|
|— dataset/
|   |— hdi_dataset.csv
|
|— model/
|   |— train_model.py
|   |— predict_hdi.py
|   |— hdi_model.pkl
|
|— templates/
|   |— index.html
|   |— login.html
|   |— dashboard.html
|   |— result.html
|
|— static/
|   |— css/
|   |— js/
|   |— images/
|   |— charts/
|
|— database/
|   |— database.sql
|   |— schema.sql
|
|— utils/
|   |— preprocess.py
|   |— visualization.py
|   |— helper.py
|
|— reports/
|   |— prediction_reports/
|
|— tests/
|   |— test_prediction.py
|   |— test_database.py
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://localhost:5000 (Replace with deployed URL if available)

Login Credentials (Demo)	Username: admin@example.com Password: admin123
Platform / Hosting Provider	Localhost / Render / Railway (<i>choose one</i>)
Repository Link	https://github.com/golidevisriprasad/HDI-Prediction-System (Replace with your GitHub URL)
Demo Video Link	https://drive.google.com/your-demo-video (Replace with actual link)

Step 4: Run Instructions

Prerequisites

- Python 3.10 or above
- MySQL Server
- Git (optional)

Installation Steps

1. Clone the repository:

```
git clone https://github.com/your-username/HDI-Prediction-System.git
```

2. Navigate to the project folder:

```
cd HDI_Prediction_System
```

3. Install required dependencies:

```
pip install -r requirements.txt
```

4. Configure the database:
 - Create a MySQL database.
 - Import `database/schema.sql`.
5. Configure environment variables:
 - Copy `.env.example` to `.env`.
 - Update database username, password, and host.
6. Run the application:

```
python app.py
```

7. Open your browser and access:

`http://localhost:5000`

8. Log in using the demo credentials and test the HDI prediction features.
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Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy depends on the quality of the dataset.	Use updated and validated HDI datasets.
2	Internet connection is required for cloud deployment.	Run the application locally when offline.
3	Live integration with UNDP APIs is not implemented.	Import updated datasets manually in CSV format.

Executable Files Summary

The **Human Development Index (HDI) Prediction System** includes all essential executable and deployment files required to run the application successfully. The project contains a complete source code repository, machine learning model, datasets, database scripts, frontend and backend modules, dependency files, documentation, and test cases. Clear setup instructions and deployment details enable evaluators to install, execute, and verify the application with minimal configuration, ensuring a smooth evaluation process.